

IOS-MCN releases open source Agartala 0.4.0 to enable private 5G

The Indian Open Source for Mobile Communication Networks (IOS-MCN) Consortium has launched Agartala 0.4.0, an open source software platform that enables enterprises and institutions to build and operate private 5G networks, marking a key step towards deployable, industry-grade 5G infrastructure developed in India.

Funded by the Ministry of Electronics and Information Technology (MeitY), Government of India, the platform is designed to support pilot deployments that can pave the way for large-scale adoption of private 5G networks across sectors.

The IOS-MCN Consortium is led by the Indian Institute of Science (IISc), Bengaluru, in collaboration with the Indian Institute of Technology (IIT), Delhi, and the Centre for Development of Advanced Computing (C-DAC). The initiative focuses on translating research-led innovation into operational national infrastructure through publicly funded open source development.

Agartala 0.4.0 is designed for factories, campuses, research institutions and startups, offering faster, more reliable and secure connectivity compared to Wi-Fi or public mobile networks, while remaining more cost-effective. The release positions open source as a strategic foundation for India's private 5G ecosystem, reducing dependence on proprietary telecom platforms.



indigenous infrastructure. The system now runs exclusively on Indian cloud and GPU resources, with all language datasets, models and citizen interactions confined within Indian jurisdiction, ensuring complete data sovereignty.

The milestone aligns directly with IndiaAI Mission objectives and signals a decisive shift towards self-reliant, scalable sovereign AI systems capable of operating without foreign hyperscalers.

At the core of the migration is an open source, cloud-agnostic and interoperable architecture designed to ensure vendor neutrality and long-term strategic autonomy. The framework is modular and reusable, enabling ministries, public sector units and national programmes to replicate hyperscaler-to-Indian-cloud transitions.

Operational metrics underline the shift's viability. The deployment recorded up to 40 percent performance gains, 20–30 percent cost savings, 99.99 percent uptime and zero data loss while migrating more than 200 TiB of data and over 3.5 billion files. Executed within two to three months, the transition covered BHASHINI's complete AI stack, including datasets, models, APIs, containerised services, orchestration pipelines, databases and storage.

Sunil Gupta, co-founder, managing director and CEO, Yotta Data Services, said, "Yotta's successful deployment of BHASHINI on Shakti cloud marks a defining moment for India's data sovereignty journey. This transition highlights that hyperscale, mission-critical AI platforms can be built and operated entirely on sovereign infrastructure, without compromise."

NVIDIA open sources its Alpamayo AI model to help cars think in chaotic traffic

NVIDIA has open sourced its Alpamayo family of reasoning-led AI models, along with simulation tools and datasets, aiming to accelerate autonomous vehicle



Jensen Huang, CEO, NVIDIA

development and push self-driving beyond controlled test environments into chaotic real-world roads.

The new stack introduces what the company calls a "thinking" approach, enabling vehicles to reason through unfamiliar situations rather than rely mainly on rules or large-scale pattern matching. NVIDIA says this step-by-step, human-like reasoning is critical for markets such as

India, where heterogeneous traffic, weak lane discipline, unpredictable behaviour and complex road conditions have long challenged autonomy.

"The Alpamayo technology is based on a revolutionary new approach called 'thinking'. For the very first time, we have added to the car the ability to reason... the way we humans do," said Jensen Huang, CEO, NVIDIA.

Positioned as the world's first reasoning-based autonomous driving AI stack, Alpamayo shifts development away from closed, proprietary systems towards broader developer participation, faster experimentation and quicker localisation.

Mercedes-Benz has already integrated NVIDIA's platform into production vehicles. Its latest-generation S-Class runs the NVIDIA DRIVE AV Level-4 full-stack system, marking a five-year collaboration between the two companies.

"With our Alpamayo technology, Mercedes is among the safest autonomous platforms in the world. The next one is Tesla, and almost everybody else is quite far behind," Huang said.